

Search for Planetary-mass Black Holes with an Improved Viterbi Algorithm

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Primordial black holes in the planetary-mass range have attracted renewed interest; however, the search for gravitational waves from such binaries remains challenging due to their long-lived nature. In this work, we present, define, and validate a fully operational search pipeline developed to detect planetary-mass binaries during their inspiral phase. We use the Viterbi algorithm, a dynamic programming technique that recovers the most likely track based on a Hidden Markov Model. To enhance its performance, we introduce a novel time-frequency representation of the data and a candidate isolation procedure that separates signals from background noise. The evaluation of candidates is carried out using the two detection statistics, n_σ and NMSE, which quantify the power significance and the consistency with the expected binary evolution. We then validate the search method using O3 LIGO Hanford data with a population of injected signals. The pipeline is able to recover most of the signals with a fixed false-alarm ratio of 3%, covering Galactic scales across most of the parameter space and reaching luminosity distances $\gtrsim 100$ kpc in the most sensitive region. For each candidate, we also obtain an estimate of the system's chirp mass, whose accuracy remains high throughout the detectable range, enabling a rapid characterization of the system upon detection.

I. INTRODUCTION

A new era in the study of black holes emerged with the discovery of gravitational waves (GWs) [1]. In particular, the routine detection of binary black hole mergers [2–7] is helping to characterize these bodies in a comprehensive way. This is one of the reasons for the renewed interest in the long-standing hypothesis of primordial black holes (PBHs). Such bodies would be formed in the early Universe, and would be able to explain some of the most intriguing questions of the current cosmological model [8–10]. One of the most interesting aspects is that an specific mass distribution of PBHs could account for all or a fraction of the intriguing dark matter (DM) [11, 12]. Notice that we talk about a mass distribution, since different theories behind PBHs formation predict a vast range of possible masses [13].

With the recent increase in popularity of PBHs [14], and the hundreds of binary black hole mergers already detected [7], the community is pointing to GWs as the promising channel to search for this DM candidate [15, 16]. Specifically, the discovery of a sub-solar mass (SSM) black hole would be an unambiguous evidence of the existence of PBHs.

Recent results by the LIGO-Virgo-KAGRA (LVK) collaboration; as low-spin black holes [7, 17], compromised mass gaps [3, 4, 18] or SSM candidates [19, 20], have motivated the search of PBHs. However, state-of-the-art

search methods cannot cover the complicated mass range below the sub-solar, where long inspiral signals become computational prohibitive for matched filtering [21]. An increasing number of templates with enormous durations limits matched filtering to smallest masses of $0.1M_\odot$.

When the component masses are sufficiently small, the search can be addressed using continuous waves (CWs) search methods. In essence, for such light systems, the GW signal during the inspiral phase becomes similar to a CW. Constraints in the asteroid-mass range have been reported using these CW based methods [22–24]; however, their applicability is limited by a maximum allowed spin-up of $\dot{f} < 10^{-8} \text{ Hz s}^{-1}$, which corresponds to chirp masses less than $\mathcal{O}(10^{-5}) M_\odot$.

Recently, the intermediate mass window has attracted attention due to some observational evidences. The Subaru HSC observations of M31 and the five-year OGLE microlensing survey toward the Galactic bulge have identified ultra-short microlensing events consistent with planetary-mass PBHs [25, 26]. Moreover, the NANOGrav collaboration found evidence of a nano-Hertz stochastic GW background using pulsar timing arrays [27], which may be linked to scalar-induced GWs associated with PBHs in the planetary-to-subsolar mass range [28, 29].

In this paper, we aim to investigate the uncovered mass range between sub-solar CBC and CWs search methods, mainly $10^{-4}M_\odot \lesssim M_{\text{PBH}} \lesssim 10^{-1}M_\odot$. We propose here a fully operational detection method, based on the Viterbi algorithm. Our search is still a CWs technique in essence, but specifically tuned for a “transient” continuous signal, as the proper low-mass inspiral.

The pioneering work exploring part of this mass range in GW data [22, 24, 30] was carried out using the *Generalized Frequency-Hough* algorithm, a standard technique in CW

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searches. An alternative approach, the Band-Sampled Data COmpact Binary Inspiral (BSD-COBI) method [31], has also been proposed, based on heterodyning techniques to track the signal evolution.

In contrast, our study employs the Viterbi dynamic programming algorithm [32], which infers the most probable sequence of hidden Markov states in a model-agnostic framework. The Viterbi algorithm has already been successfully applied in CWs [23, 33–36], mainly through the SOAP package [34], a fully operational pipeline that extracts the most likely hidden signal from noisy GW data.

The Viterbi algorithm is computationally efficient, as it operates in a model-agnostic framework without requiring a precomputed bank of candidate tracks. Due to its template-free nature, the algorithm remains sensitive to signals with more complex waveform features, including higher-order corrections or additional physical effects such as dark-matter environments.

A recent work [37] demonstrated the effectiveness of the Viterbi algorithm in searching for planetary-mass PBH inspirals assuming Gaussian noise. Here, we extend this approach by developing a fully operational search pipeline tailored to this mass range. We include a new time-frequency domain representation of the data, improving the accuracy of Viterbi. Moreover, we develop a candidate isolation method which discriminates potential signals from background noise. In addition, we introduce newly detection statistics that improve the significance of the search and allow us to distinguish true signals from noise fluctuations. The resulting pipeline enables end-to-end searches on real data, from the strain time series to candidate evaluation.

The paper is structured as follows. In Sec. II, we describe the key elements of the search, including the properties of long inspiral signals, the Viterbi framework adopted in this work, the adaptation of short Fourier transforms (SFTs) to this class of signals, and the detection statistic used to identify significant candidates. In Sec. III, we describe the full pipeline architecture and assess its performance on a search using LIGO O3b data. Finally, in Sec. IV, we summarize our results and discuss prospects for future searches in new observing runs.

II. METHODOLOGY

A. Long Inspirals

When two compact objects start inspiraling, orbital energy is radiated away via GW emission leading to an increasing ω_{orb} , and so, an increasing GW frequency, f_{gw} . Equating the power dissipated as a result of gravitational wave emission with the rate of change of the orbital energy of the system [38], we find the spin-up:

$$\dot{f}_{\text{gw}} = \frac{96}{5} \pi^{8/3} \left(\frac{GM}{c^3} \right)^{5/3} f_{\text{gw}}^{11/3}, \quad (1)$$

where $\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}$ is the chirp mass of the system. For the most massive PBHs considered here, the f_{gw} at the inner most stable circular orbit (ISCO) reaches $\mathcal{O}(\text{MHz})$, far away from the detectable frequency band of current detectors. This guarantees that the circular orbit approximation remains valid throughout the entire frequency band, making this equation applicable across the relevant observational range. Integrating Eq. (1), we find the GW frequency evolution as function of time:

$$f_{\text{gw}}(t) = \frac{1}{\pi} \left(\frac{5}{256} \right)^{3/8} \left(\frac{GM}{c^3} \right)^{-5/8} (t_{\text{coal}} - t)^{-3/8}. \quad (2)$$

We identify that f_{gw} formally diverges at a finite value of time, t_{coal} , the *coalescence time*. Then, strictly speaking, the frequency evolution is a function of the *time to coalescence*, $(t_{\text{coal}} - t)$, i.e., the remaining time until the merger. Equivalently, we can write

$$(t_{\text{coal}} - t) = \left(\frac{5}{256} \right) \left(\frac{GM}{c^3} \right)^{-5/3} (\pi f_{\text{gw}})^{-8/3}. \quad (3)$$

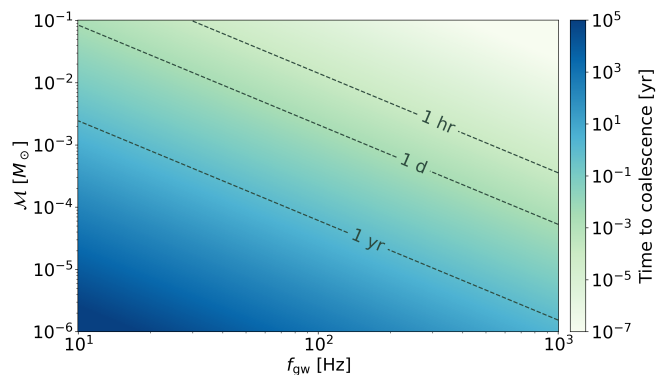


FIG. 1. Time to coalescence as a function of the chirp mass and frequency of the system. The broad range of signal durations across the parameter space indicates that distinct techniques are required to effectively investigate the presence of PBHs at different mass scales.

Due to their small masses, planetary-mass PBHs are expected to undergo inspirals that last significantly longer in the detectable frequency band than typical BBHs detected by the LVK collaboration, potentially spanning months or even years [30]. This can be easily visualized by representing the *time to coalescence* as a function of the chirp mass of the binary and the GW frequency. In Fig. (1), we illustrate that PBHs inspirals are indeed ideal candidates for CWs and transient CWs search methods.

A novel aspect of this work is that we will not operate on conventional time–frequency maps. In standard spectrograms, the inspiral signal traces a characteristic chirp pattern (see Fig. 2, top panel), whose frequency evolution is described by Eq. (1), and can be re-written as:

$$\frac{\dot{f}_{\text{gw}}}{f_{\text{gw}}^{11/3}} = C \cdot \mathcal{M}^{5/3}, \quad (4)$$

where C is a constant value. But, here one can introduce a coordinate change as follows:

$$\frac{d}{dt} \left(f_{\text{gw}}^{-8/3} \right) = K \cdot \mathcal{M}^{5/3}, \quad (5)$$

where K is a constant value:

$$K = -\frac{256}{5} \pi^{8/3} \left(\frac{G}{c^3} \right)^{5/3}. \quad (6)$$

In Eq. (5) we just have one independent variable, the chirp mass. So, we lead to a new frequency map $(t, f^{-8/3})$ where the long inspirals become straight lines (Fig. 2). Furthermore, in these transformed coordinates, the frequency evolution depends only on the chirp mass of the system. Both features play a key role in improving the performance of the search method.

B. The Viterbi Algorithm

The Viterbi algorithm is a dynamic programming algorithm that allows to infer the most probable sequence of hidden states in a Markov model based on noisy data. It operates within the framework of Hidden Markov Models, where the true state of the system evolves according to a Markov process, but is not directly observable. In this study, we use the Viterbi algorithm as a novel technique for searching long-lasting inspirals, following the idea proposed in [37]. The approach is implemented through the SOAP package [34–36, 39], which has been originally developed to search for CWs signals.

In the SOAP implementation, the input frequency time-series data is divided into N segments of equal duration [34], forming the dataset of time series segments $\mathbf{x}_i$, which we denote as $D \equiv \{\mathbf{x}_i\}$, where each index i corresponds to a specific time segment. The track we aim to recover is the sequence of signal frequencies $\boldsymbol{\nu} \equiv \{\nu_i\}$, where ν_i represents the GW frequency within segment $\mathbf{x}_i$. The goal is to evaluate all possible frequency tracks and identify the one that maximizes the posterior probability, i.e. the most likely signal path given the data:

$$\hat{\boldsymbol{\nu}} = \arg \max_{\boldsymbol{\nu}} P(\boldsymbol{\nu} | D). \quad (7)$$

Following Bayes' theorem, the posterior probability takes the form:

$$p(\boldsymbol{\nu} | D) = \frac{p(\boldsymbol{\nu}) p(D | \boldsymbol{\nu})}{p(D)}, \quad (8)$$

where $p(\boldsymbol{\nu})$ is the prior probability of the track, $p(D | \boldsymbol{\nu})$ is the observation likelihood (the probability of the data given the track) and $p(D)$ is the model evidence. Notice that $P(D)$ is constant for all $\boldsymbol{\nu}$, so maximizing $P(\boldsymbol{\nu} | D)$ is equivalent to just maximizing the numerator, i.e. the joint probability $P(\boldsymbol{\nu}, D)$.

We can expand explicitly the joint probability and apply the observational independence and the Markov property

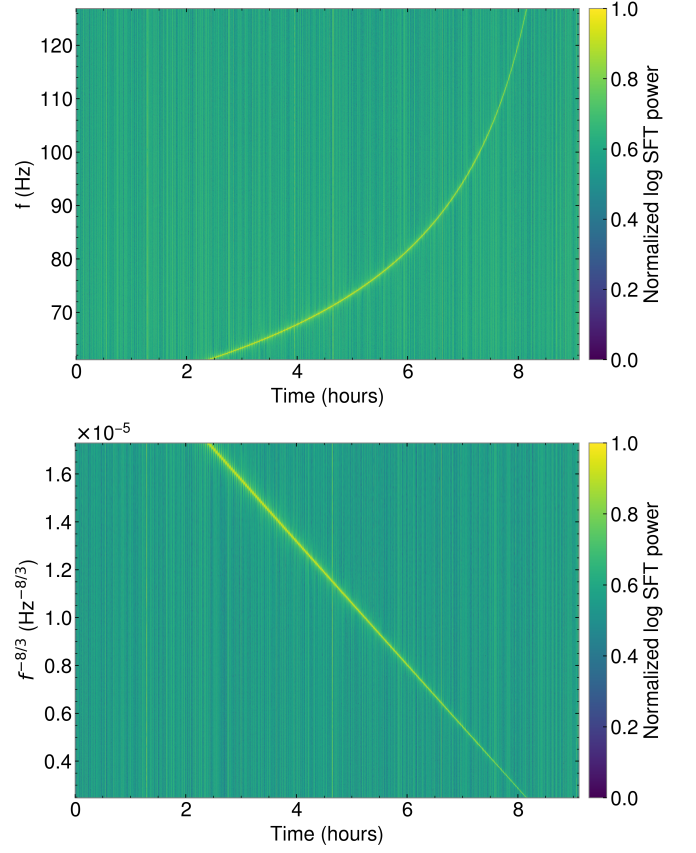


FIG. 2. Time-frequency representation of O3b LIGO Hanford data with a high-SNR signal injected. Top panel: Spectrogram of a strain time series with a visible chirp of a long inspiral. Bottom panel: New frequency map ($f^{-8/3}$) of the strain, where the characteristic chirp becomes a straight line.

[40], leading to

$$P(\boldsymbol{\nu}, D) = P(\nu_0) \prod_{n=1}^N P(\nu_n | \nu_{n-1}) \prod_{n=0}^N P(x_n | \nu_n), \quad (9)$$

where $P(\nu_n | \nu_{n-1})$ is the “transition” probability for ν_n given the frequency at the last step was ν_{n-1} . Now, for numerical stability, we work in the logarithmic domain. Therefore, the most probable signal track $\hat{\boldsymbol{\nu}}$ that maximizes the posterior probability is then found by maximizing the logarithm:

$$\hat{\boldsymbol{\nu}} = \arg \max_{\boldsymbol{\nu}} [\log P(\boldsymbol{\nu}, D)] \quad (10)$$

$$= \arg \max_{\boldsymbol{\nu}} \left\{ \log P(\nu_0) + \log P(\mathbf{x}_0 | \nu_0) + \sum_{i=1}^N [\log P(\nu_i | \nu_{i-1}) + \log P(\mathbf{x}_i | \nu_i)] \right\}. \quad (11)$$

Rather than evaluating every path, it recursively computes the maximum probability of arriving at each state at each time step, keeping track of the most likely predecessor, drastically reducing the search space. In fact, for a

state space of size S and a sequence of length N , a naive search would require evaluating $\mathcal{O}(S^N)$ possibilities, while Viterbi achieves the same result with $\mathcal{O}(S^2 N)$ operations [40].

The transition matrix T encodes the prior log-probabilities $\log P(\nu_n | \nu_{n-1})$, which represent the probability of the signal's frequency transitioning from the $(n-1)$ th to the (n) th state. It provides a natural framework for imposing loose model constraints. For instance, some recent works [34, 37] restricted successfully the track to move at most one bin per step: up, center, or down (UCD) transition or “jump”, reducing T to three distinct values. This configuration was chosen based on the expected smoothness of the track. As we will see later, we will take advantage of the transition matrix to improve Viterbi's performing.

C. Short-time Fourier Transform

The primary input to the Viterbi algorithm is the set of Short-Time Fourier Transforms (SFTs) computed from the strain time-series data frames. These are obtained by dividing the full data stream $s[n]$, where n indexes discrete time samples, into segments of duration T_{SFT} , applying a window function $w[n]$, and performing a Discrete Fourier Transform on each segment. This process yields a time-frequency representation $\tilde{s}[n, k]$, where k labels frequency bins. The frequency resolution is directly define as $\Delta f = 1/T_{\text{SFT}}$, while the total number of SFTs is given by $N_{\text{SFT}} = T/T_{\text{SFT}}$, where T the total observation time.

In the context of CW and transient-CW search techniques, it is typically assumed that the signal frequency remains confined within a single frequency bin during each SFT. For a given SFT duration and an inspiral signal from a binary system with chirp mass $\mathcal{M}$, the maximum frequency f_* that satisfies the condition [37] that the signal remains within the same frequency bin is given by

$$f_* = \frac{1}{\pi} \left(\frac{5\pi}{96} \right)^{3/11} \left(\frac{G\mathcal{M}}{c^3} \right)^{-5/11} T_{\text{SFT}}^{-6/11}. \quad (12)$$

Above this frequency, the frequency evolution becomes sufficiently rapid as the system approaches the merger phase, causing the signal to drift out of a frequency bin.

Eq. (12) indicates that shorter SFT durations allow access to higher maximum frequencies, enabling the inclusion of more rapidly evolving signals and thus potentially increasing the sensitivity. On the other hand, reducing the SFT duration increases the number of segments, and hence the number of degrees of freedom. As a consequence, the statistical fluctuations of the detection statistic become larger, reducing its statistical significance. These two competing effects imply the existence of an optimal choice of SFT duration (or equivalently, an optimal maximum frequency) that balances sensitivity to high-frequency signals and statistical stability.

The optimal SFT duration was derived in [37] by maximizing the detection horizon distance. It was shown [37] that the optimal frequency range can be obtained by maximizing the following function:

$$F(f_0, f_*) = \frac{f_0^{2/3}}{f_*^{11/24}} \sqrt{\int_{f_0}^{f_*} \frac{df}{f^{7/3} S_n(f)}}. \quad (13)$$

Here, f_0 and f_* denote the low- and high-frequency cutoffs, respectively. Note that, by construction, the shape of Eq. (13) is similar to the signal-to-noise ratio (SNR). This function is determined solely by the shape of the noise power spectral density (PSD) and does not depend on the signal parameters, such as the chirp mass. For the LIGO Handford O3 PSD [41], we find that the function is maximized at $f_{0,\text{opt}} = 61.1$ Hz and $f_{*,\text{opt}} = 126.8$ Hz¹.

Using Eq. (12) and normalizing it with respect to the optimal high-frequency cutoff $f_{*,\text{opt}}$, we obtain the optimal SFT duration for the O3 dataset:

$$\begin{aligned} T_{\text{SFT}}^{\text{opt}} &= \left(\frac{5\pi}{96} \right)^{1/2} \left(\frac{G\mathcal{M}}{c^3} \right)^{-5/6} (\pi f_{*,\text{opt}})^{-11/6} \\ &= 8.50 \text{ s} \left(\frac{\mathcal{M}}{10^{-2} M_\odot} \right)^{-5/6} \left(\frac{f_{*,\text{opt}}}{126.8 \text{ Hz}} \right)^{-11/6}. \end{aligned} \quad (14)$$

See Ref. [37] for a more detailed derivation.

As the optimum SFT duration depends on the chirp mass of the system we are searching, covering a mass range in our search means using a range of T_{SFT} . This turns the optimum SFT duration into one of the unknown variables, directly related with the binary's chirp mass.

D. Detection Statistics

In order to identify significant candidates, we require a detection statistic capable of quantifying how unlikely the recovered Viterbi track is under the noise-only hypothesis. Since the algorithm selects the most probable path in the time-frequency map, the relevant quantity is not the power accumulated along an arbitrary track, but the maximum accumulated power among all the tracks explored by Viterbi. Following a previous work [37], we use the statistic n_σ as one of the main ranking quantities of the search.

Let us consider a candidate track $\nu = \{\nu_i\}$, where the index i labels the SFT segment. For each point of the track, we define the matched-filter-like signal-to-noise ratio (SNR) contribution ρ_i^{mf} , and the total incoherent SNR accumulated along the track as

$$\rho_{\text{tot}}^2(\nu) = \sum_{i=1}^{N_{\text{SFT}}} |\rho_i^{\text{mf}}(\nu_i)|^2. \quad (15)$$

¹ For the LIGO O5 PSD the optimum frequency band lies in a similar region, with $f_{0,\text{opt}} = 57.5$ Hz and $f_{*,\text{opt}} = 134.4$ Hz.

The Viterbi algorithm returns the track that maximizes the posterior probability, and therefore the relevant quantity for candidate selection is

$$\rho_{\text{tot,max}}^2 = \max_{\nu} \rho_{\text{tot}}^2(\nu). \quad (16)$$

In the absence of a signal, $\rho_{\text{tot,max}}^2$ does not follow the same distribution as the accumulated power of a single fixed track, since Viterbi maximizes over a large number of correlated paths. This effect must be included in the definition of the detection statistic.

We then define n_σ as the number of standard deviations by which the recovered Viterbi SNR differs from the expected noise-only background,

$$n_\sigma = \frac{\rho_{\text{tot,max}}^2 - \mu(\rho_{\text{tot}}^{\text{opt}} = 0, N_{\text{SFT}})}{\sigma(\rho_{\text{tot}}^{\text{opt}} = 0, N_{\text{SFT}})}. \quad (17)$$

Here, $\mu(\rho_{\text{tot}}^{\text{opt}} = 0, N_{\text{SFT}})$ and $\sigma(\rho_{\text{tot}}^{\text{opt}} = 0, N_{\text{SFT}})$ are, respectively, the mean and standard deviation of the noise-only distribution of $\rho_{\text{tot,max}}^2$, for a given number of SFTs. The condition $\rho_{\text{tot}}^{\text{opt}} = 0$ denotes the absence of an injected signal. In practice, these quantities encode the background expected from random noise fluctuations after the Viterbi maximization.

This statistic provides an intuitive measure of the significance of a recovered track. A candidate with large n_σ corresponds to a path whose accumulated power is far from the typical noise-only maximum, and is therefore less likely to be produced by a random fluctuation. This is particularly useful in our search, since each choice of T_{SFT} produces a different number of SFTs, and hence a different noise background. The dependency on N_{SFT} in Eq. (17) allows us to compare candidates obtained from different time–frequency representations.

For a sufficiently loud signal, the Viterbi path is expected to coincide with the true signal track. In this limit, the mean accumulated SNR along the recovered track can be approximated by the mean of the non-central χ^2 distribution [37], which is defined as

$$\langle \rho_{\text{tot,max}}^2 \rangle \simeq 2N_{\text{SFT}} + (\rho_{\text{tot}}^{\text{opt}})^2, \quad (18)$$

where $\rho_{\text{tot}}^{\text{opt}}$ is the optimal total SNR of the signal. Substituting this expression into Eq. (17), we obtain the analytical approximation

$$\langle n_\sigma \rangle \simeq \frac{2N_{\text{SFT}} + (\rho_{\text{tot}}^{\text{opt}})^2 - \mu(\rho_{\text{tot}}^{\text{opt}} = 0, N_{\text{SFT}})}{\sigma(\rho_{\text{tot}}^{\text{opt}} = 0, N_{\text{SFT}})}. \quad (19)$$

This expression corresponds to the large-SNR limit in which the recovered Viterbi track follows the physical signal. As the signal becomes fainter, the loud signal assumption breaks down, and this approximation does so, since the algorithm can instead lock onto a noise-dominated path. In Fig. 3 we show the evolution of this metric as a function of the luminosity distance, for experimental values (dots) and the analytical approximation (line).

One can detect how the approximation breaks once the signal enters the low-SNR regime. Nevertheless, Eq. (19) provides the theoretical basis for interpreting n_σ as a significance-like statistic and motivates its use as one of the main quantities for ranking candidates in this work.

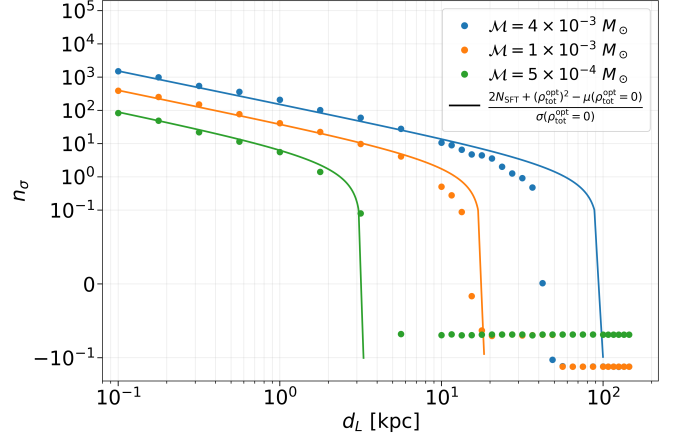


FIG. 3. Evolution of the detection statistic n_σ with luminosity distance. Points show the values recovered from O3 LIGO Handford data with injected signals, computed using Eq. (17). The solid line corresponds to the analytical expected value of Eq. (19). At larger distances, the injected signals become weaker, the loud signal assumption breaks down and n_σ approaches the noise-dominated regime.

In addition to the power-based statistic n_σ , we introduce a second statistic aimed at quantifying the morphological consistency of the recovered track with the expected frequency evolution of a compact-binary inspiral. This is particularly important in the low-SNR regime, where the accumulated power along the Viterbi path can be comparable to ordinary noise fluctuations. In this case, n_σ alone may not be sufficient to distinguish a real slowly evolving signal from a noise-induced track.

For this purpose, we use the normalized mean square error (NMSE). Let

$$\hat{\mathbf{y}} = \{\hat{y}_i\}_{i=1}^{N_{\text{seg}}} \quad (20)$$

be the portion of the Viterbi track selected as a candidate, written in the remapped frequency coordinate $y \equiv f^{-8/3}$. Here, N_{seg} denotes the number of time bins in the candidate segment. From Eq. (5), the expected inspiral evolution in this coordinate is approximately linear,

$$y_{\text{model}}(t, \mathcal{M}) = K \cdot \mathcal{M}^{5/3}(t - t_0), \quad (21)$$

where K is the constant introduced in Eq. (6), $\mathcal{M}$ is the chirp mass, and t_0 is the initial time of the candidate track. Therefore, in the transformed map, fitting the candidate morphology reduces to testing whether the recovered Viterbi track is compatible with a “straight line” whose slope is determined by the chirp mass. Note that this is not exactly true, as Doppler modulations imply that the frequency evolution is not exactly a line, but for

the short T_{SFT} of this work, we can approximate it as a line.

For a given value of the chirp mass, we define the NMSE as

$$\text{NMSE} = \frac{\sum_{i=1}^{N_{\text{seg}}} [\hat{y}_i - y_{\text{model}}(t_i, \mathcal{M})]^2}{\sum_{i=1}^{N_{\text{seg}}} \hat{y}_i^2}. \quad (22)$$

The normalization by the total squared amplitude of the recovered track makes the statistic dimensionless and allows us to compare candidates located in different frequency regions. Small NMSE values indicate that the track follows the expected inspiral morphology, whereas large NMSE values correspond to tracks that are poorly described by the compact-binary frequency evolution.

Since the chirp mass of the source is not known *a priori*, we minimize the NMSE over the target mass range,

$$\text{NMSE}_{\min} = \min_{\mathcal{M}} (\text{NMSE}), \quad (23)$$

with

$$\mathcal{M} \in [10^{-4}, 10^{-1}] M_{\odot}. \quad (24)$$

The value of the chirp mass that minimizes the statistic,

$$\hat{\mathcal{M}} = \arg \min_{\mathcal{M}} \text{NMSE}(\mathcal{M}), \quad (25)$$

provides a first estimate of the source chirp mass. We do not interpret this value as a precise parameter-estimation result, since the Viterbi track is obtained from a semi-coherent search and may be affected by noise features, spectral lines, or imperfect candidate isolation. However, it provides a useful consistency check: signal-like tracks should not only have an excess of accumulated power, but should also be well fitted by an inspiral trajectory within the physical mass range explored by the search.

The two statistics are therefore complementary. The quantity n_{σ} measures how significant the recovered track is with respect to the noise-only Viterbi background, while $\text{NMSE}_{\min}$ (NMSE, from now on) measures how compatible the track is with the expected long-inspiral morphology. A noise fluctuation can occasionally produce a large accumulated power, but it is not expected to follow the specific linear behaviour in the $(t, f^{-8/3})$ plane. On the other hand, a low-SNR signal may not produce a very large n_{σ} , but can still be identified through a small NMSE if its recovered track follows the expected inspiral evolution. For this reason, we evaluate candidates in the two-dimensional detection-statistic plane $(n_{\sigma}, \text{NMSE})$, which allows us to combine statistical significance and physical consistency in a single candidate-selection criterion.

III. APPLICATION TO LVK DATA

In this work, we present a fully operational search pipeline, end-to-end, from time-series strain to a candidate

evaluation. It is designed to detect transient CWs, produced during the inspiral phase of PBHs binaries within the planetary-to-sub-solar mass range.

Based on the Viterbi algorithm, this signal-agnostic method is capable of recovering faint, slowly evolving signals buried in interferometric data. The main challenge of this search lies in the nature of the signal itself. As discussed above, these long-duration inspirals differ both from typical CBCs and from standard CW signals. In this intermediate regime, between short-duration transients and nearly monochromatic signals, the waveform evolution strongly depends on the binary parameters.

As a result, the low-mass part of this range can produce signals with minimal frequency evolution within our frequency band over timescales of $\mathcal{O}(\text{moths})$, whereas higher-mass systems generate more rapidly evolving tracks that sweep across the band on timescales of $\mathcal{O}(\text{minutes})$. To successfully isolate such signals from random noise, a search method must be both robust and flexible enough to accommodate this wide diversity of signal behaviours across the mass range.

In this section, we first describe the multi-stage structure of the pipeline, from the transformation of the strain data into the frequency domain to the identification of significant candidates. We then apply the pipeline to real data from the third observing run (O3) of LIGO Handford, validating its performance and assessing its sensitivity through a controlled scenario with injected signals.

A. Pipeline Architecture

The search method presented here is built around the Viterbi algorithm. However, Viterbi constitutes only one component within a broader multi-stage procedure.

The pipeline begins by transforming the interferometer time-series into short-time Fourier transforms (SFTs), as the full analysis is carried out in the frequency domain.

At this stage, the key parameter is the coherence time of the Fourier transform, T_{SFT} . As seen in Sec. II C, a previous work [37] has shown that the optimal $T_{\text{SFT}}^{\text{opt}}$ for signal recovery depends directly on the chirp mass of the system. Since the signal parameters are not known *a priori* and our search targets a broad mass range, we consider a set of T_{SFT} values rather than a single choice.

We determine the optimal T_{SFT} values across the mass range $10^{-4} M_{\odot} \lesssim \mathcal{M} \lesssim 10^{-1} M_{\odot}$ by requiring the SNR loss, relative to the maximum achievable SNR, to remain below 1% [37]. As a result, the data are represented by multiple time-frequency maps rather than by a single spectrogram. This step constitutes the computational bottleneck of the pipeline, although can be easily reduced as we will discuss at the end of this section.

Once the spectrograms are built, following the variable change presented in Sec. II A, we remap the frequency representation leading to a new frequency map $(t, f^{-8/3})$, where the chirps become straight lines (Fig. 2).

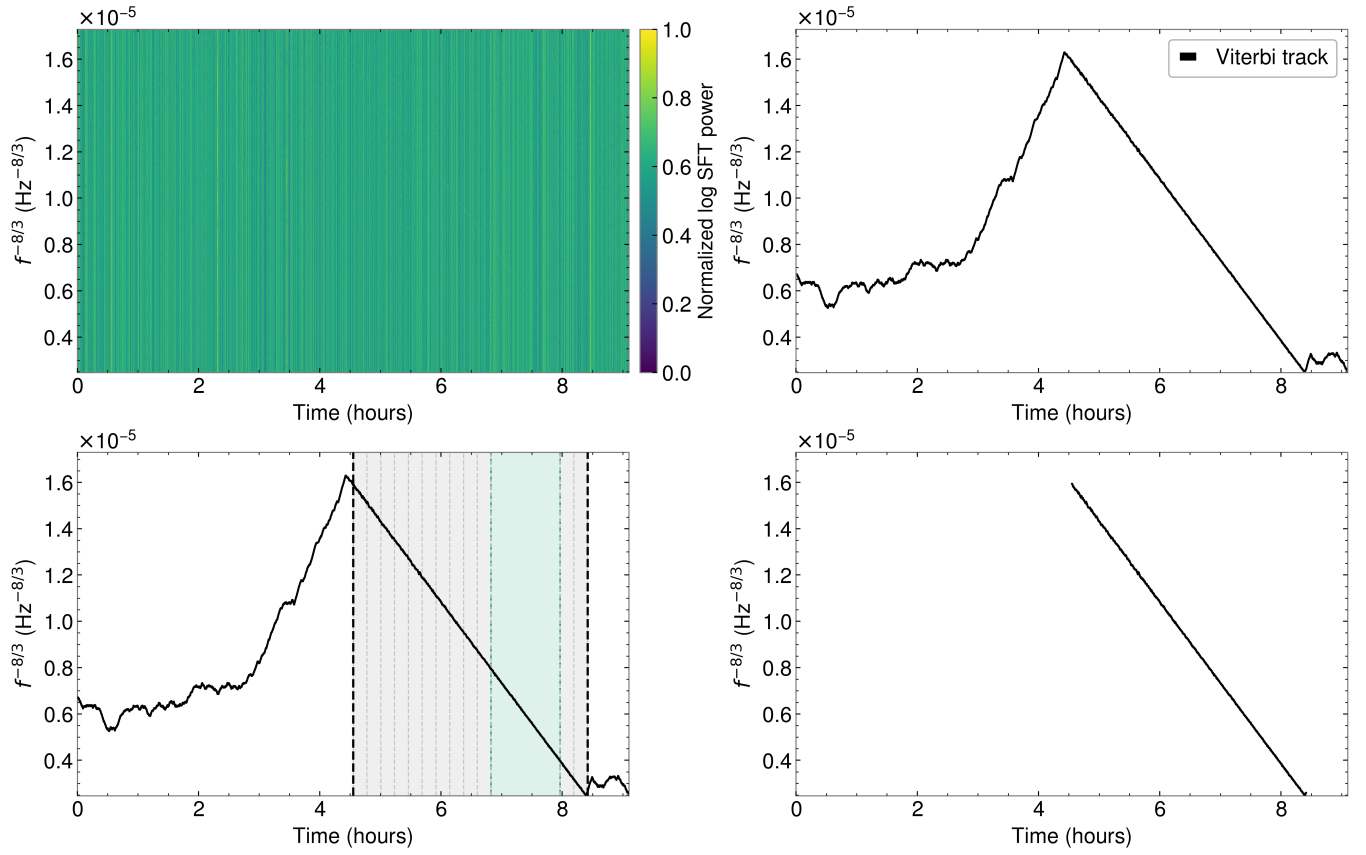


FIG. 4. Phases of the search pipeline applied to an injected long-inspiral signal. From top left to bottom right, we visualize the different stages of the pipeline. Top left: the remapped frequency representation introduced in Sec. II A. Top right: we recover the Viterbi track, the most likely frequency path given the data. Bottom left: isolation procedure, where it first recovers the most power-significant window (green) and then expand it using small windows to both sides. Bottom right: the final candidate, associated with a pair of ranking statistics (n_σ , NMSE). The injected signal corresponds to $[10^{-2} M_\odot, 80 \text{ kpc}]$.

This change is motivated for two main reasons. The first one is that now we can take advantage of the transition matrix (see Sec. II B) to create a prior to center and down (CD) jumps, so Viterbi can easier follow the signal. The second reason lies in the relation with the chirp mass. With this new map, fitting the slope is just a function of the system's chirp mass, which lays the foundation for using the fitting detection metric NMSE (see Sec. II D), very useful to recognize significant candidates.

Once the spectrograms are built, following the variable change presented in Sec. II A, we remap the frequency representation leading to a new frequency map $(t, f^{-8/3})$, where the chirps become straight lines (Fig. 2). This change is motivated for two main reasons. The first one is that now we can take advantage of the transition matrix (see Sec. II B) to create a prior to center and down (CD) jumps, so Viterbi can easier follow the signal. The second reason lies in the relation with the chirp mass. With this new map, fitting the slope is just a function of the system's chirp mass, which lays the foundation for using the fitting detection metric NMSE (see Sec. II D), very useful to recognize significant candidates.

Now, with a set of remapped spectrograms constructed, the Viterbi algorithm is applied independently to each of

them, exploiting its computational efficiency. For each spectrogram, the algorithm identifies the most probable track given the data. This allows us to select the optimal T_{SFT} representation by comparing the corresponding detection statistic, n_σ (see Sec. II D), as the spectrogram that best matches the signal will yield the track with the highest accumulated power.

The Viterbi track has been recovered, the next step is to identify the portion of the track that is most likely associated with a physical signal. To this end, we implement a four-stage isolation procedure. First, the recovered track is divided into eight time windows. For each window, we compute the fraction of the total accumulated power (ρ_i) contained within it. The two windows with the largest power fractions are retained as initial signal-candidate regions and passed to the next stage. In this second phase, the track segments of these candidate windows is compared with the expected inspiral evolution in the remapped frequency coordinate, Eq. (5). A fitting is performed over the target mass range using the normalized mean-square error (NMSE), defined in Eq. (22). We retain the window with the lowest NMSE, corresponding to the segment that best matches the expected inspiral behaviour.

The third stage refines the selected window by allowing its boundaries to expand or contract. This is done iteratively by adding or removing small portions of the track at the left and right boundaries independently, and recomputing the NMSE after each modification. This procedure allows us to recover the full signal-containing segment of the Viterbi track, as illustrated in Fig. 4, and a metric evaluation of how well it fits to the expected inspiral evolution. Finally, the isolated segment is evaluated using its detection statistics (n_σ , NMSE). A detection threshold is defined in the ranking statistics plane, using a chosen false-alarm ratio (FAR), allowing each recovered track to be classified either as consistent with noise or as a significant signal candidate.

All in all, the pipeline combines the computational efficiency of the Viterbi algorithm with a dedicated candidate-isolation procedure tailored to long inspiral signals. It enables robust searches over the broad range of planetary-to-sub-solar PBHs, while providing a physically motivated detection statistic for candidate selection in any interferometric dataset. The pipeline is available in [42], where all the codes can be implemented in any GW dataset.

B. Results

Now, we are ready to test the full pipeline on real interferometric data. Specifically, we apply the search method to data from the O3b LIGO Handford observing run, into which we inject a population of simulated long-inspiral signals. This controlled setup allows us to evaluate the behaviour of the detection statistics over a broad set of injections and to assess the sensitivity and robustness of the method.

The dataset and injection parameters used in the analysis are summarized in Table I. In total, we inject approximately 600 signals into nearly 1000 hours of interferometric data. The low-mass binary inspirals are generated using the 3.5PN TaylorT3 approximant [37, 43]. This approximant is particularly convenient for the present application because it provides the possibility of generating the signal segment by segment, which is required in practice due to memory limitations.

For each injection, we compute the time-dependent projection of the waveform onto the Hanford detector using LALSuite [44]. The resulting detector-frame signals are then added to gap-free segments of O3b strain data obtained from GWOSC public dataset [45].

Following the theoretical estimation of the optimal SFT length, see Eq. (14), we find that the chirp mass range considered here requires a set of thirteen T_{SFT} , ranging from 2 to 88 s. The SFTs are produced with the MakeSFTs routine from the LALPulsar library [44, 46]. We then use the Python soapcw package [39] to run the Viterbi algorithm, we extract the most likely tracks, and apply the full candidate-selection pipeline described in the last section.

We first run the search on the O3b raw data, which

Search parameters	
Dataset	O3b
Detector	H1
Chunk duration	32768 s
Total observation time	~ 1000 h
Sampling	512 Hz
Chirp mass, $\mathcal{M}$	$[10^{-4}, 10^{-1}] M_\odot$
Luminosity distance, d_L	$[0.1, 150]$ kpc
Mass ratio, q	1
Right ascension, α	$[0, 2\pi]$
Declination, δ	$[-\pi/2, \pi/2]$
Polarization, ψ	$[0, \pi]$
Inclination, ι	0

TABLE I. Parameters defining the benchmark search case used in this work.

provide a representative sample of noise triggers. This allows us to characterize the background distribution in the detection-statistic plane. As expected, noise triggers predominantly populate the region with large NMSE and low n_σ (see Fig. 5), corresponding to poor agreement with the inspiral model and low statistical significance.

Then, we run the pipeline after injecting the signal population into different noise realizations, as summarized in Table I. In this case, a large fraction of the recovered triggers moves towards the significant region of the plane, characterized by low NMSE and high n_σ . Furthermore, in Fig. 5, we show for each signal its corresponding SNR, computed according to Appendix A.

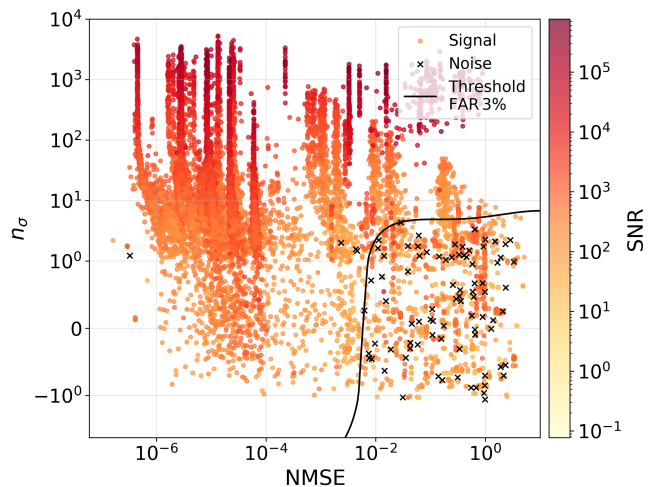


FIG. 5. Triggers distribution for the search in real data. Each search result is represented as a point in the detection-statistic plane. Dots correspond to searches performed on data containing injected signals, with the corresponding SNR, while blue crosses represent searches on pure noise realizations. The noise triggers are clearly concentrated in the lower-right region of the plane, characterized by low n_σ and high NMSE values, whereas signal candidates predominantly populate the complementary regions.

We then define a detection threshold by fixing the false-alarm ratio to $\text{FAR} = 3\%$. To this end, we construct a polynomial decision boundary in the detection-statistic plane (Fig. 5), optimized to maximize the number of true positives at fixed FAR. We consider polynomial functions up to cubic order, since higher-order boundaries do not provide a significant improvement in recovery performance.

The resulting pipeline recovers a substantial range of the injected parameter space. Nevertheless, a number of injected signals are still classified as noise. This is truly expected since there is a sensitivity limitation associated with the Viterbi reconstruction itself. Below a certain SNR, the accumulated signal power becomes comparable to, or smaller than, that of common instrumental artifacts. In this regime, the Viterbi algorithm can preferentially follow noise features or spectral lines rather than the true signal track. Additional losses arise from the candidate-isolation stage of the pipeline. Tracks that do not exhibit a well-defined concentration of power, or whose recovered morphology becomes too diffuse at low SNR, can be difficult to fit reliably with the expected inspiral evolution. These losses are expected on real data, where real noise increase the difficulty to isolate signal events.

In these low-SNR scenarios, the importance of the new metric NMSE becomes evident. While n_σ is highly effective for independently identifying high-SNR candidates, its discriminating ability get worse once the accumulated signal power becomes comparable to the background fluctuations. In this regime, the NMSE plays a key role in distinguishing inspiral-like signals from noise-induced tracks.

Now that we have run the search and defined the threshold, we can assess the sensitivity of our pipeline over the population mentioned in Table I. In Fig. 6, we show the

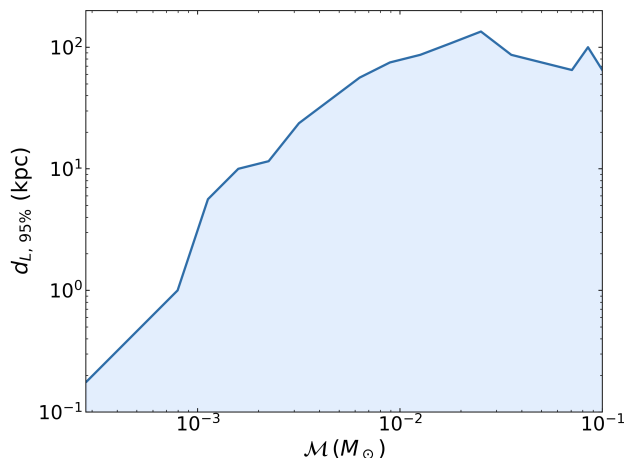


FIG. 6. Sensitivity curve at $d_{L,95\%}$. For the population injected (Table I), we can recover at least the 95% of the signals within the blue contour. As expected, the sensitivity increase with the chirp mass of the binary, up to a point where the signal evolves so fast that it becomes more difficult to isolate.

sensitivity curve. In particular, we report the $d_{L,95\%}$ curve, defined as the luminosity distance up to which signals are recovered in at least 95% of the injections.

For the injected population, we recover a substantial fraction of the signals across the explored parameter space. Above a chirp mass of $\mathcal{M} \gtrsim 5 \cdot 10^{-3} M_{\odot}$, the search reaches distances of $d_{L,95\%} \gtrsim 50$ kpc. Moreover, over most of the targeted mass range, the sensitivity extends to Galactic scales. This is particularly relevant for our science goals, as the most promising candidates are expected to be compact binaries located within the Milky Way. In the most favorable region, the pipeline is able to recover signals with a 95% efficiency out to distances of 135 kpc, according with chirp masses around $\mathcal{M} \sim 2 \cdot 10^{-2} M_{\odot}$.

An interesting trend can be observed in the high-mass range of the sensitivity curve. As the chirp mass increases, the SNR also increases, leading to the expected improvement in sensitivity. However, at the higher explored mass range, the sensitivity exhibits a slightly decrease despite the continued increase in SNR. This behaviour can be understood from the morphology of the signals. In this high mass regime, the binaries evolve significantly faster, spending only minutes within the detector band. Therefore, the signals become less similar to the long transient CWs that motivate the design of our pipeline, so the candidate recovery become more challenging. The large spin-up of these systems also limits the performance of the Viterbi reconstruction, since the transition matrix employed in this work only allows jumps of one frequency bin per step and therefore cannot fully capture such rapid frequency evolution. Nevertheless, the decrease remains moderate, and the pipeline continues to recover a large fraction of the injected population.

Finally, we evaluate the accuracy of the chirp mass estimation obtained from the NMSE minimization, Eq. (25). To quantify this rough estimate, we define the relative error as

$$\delta = \frac{\hat{\mathcal{M}} - \mathcal{M}_{\text{true}}}{\mathcal{M}_{\text{true}}}, \quad (26)$$

where $\hat{\mathcal{M}}$ is the chirp mass associated with the final candidate and $\mathcal{M}_{\text{true}}$ is the injected value. This normalization provides a dimensionless quantity that can be directly compared across the explored parameter space.

Figure 7 shows the relative error obtained for the full set of injections. Over most of the detectable region, the error remains close to zero, indicating that the pipeline is able not only to recover the signal, but also to provide a reliable estimate of its chirp mass. This behaviour closely follows the sensitivity curve shown in Fig. 6: within the region where signals are efficiently recovered, the chirp-mass reconstruction remains accurate, whereas the error rapidly increases as the injections move beyond the sensitivity limit.

We also show the marginal distributions with respect to chirp mass and luminosity distance. As expected, the relative error converges to zero throughout the high-SNR region of the parameter space and increases only for lower

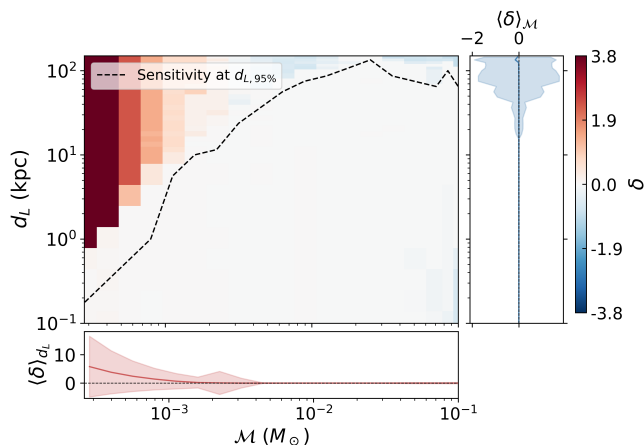


FIG. 7. Relative error of the chirp mass estimation obtained from the NMSE minimization. The central panel shows the error across the injected parameter space, while the bottom and right panels display the corresponding marginals with an standard deviation (1σ) contour. The sensitivity curve at $d_{L,95\%}$ is shown for comparison. As expected, the reconstruction remains accurate within the detectable region and progressively degrades towards lower masses and larger distances.

masses and larger distances, where the recovered tracks become progressively more affected by noise.

A small shift can be observed when comparing Fig. 6 and 7. The region characterized by low relative errors extends slightly beyond the $d_{L,95\%}$ contour. This behaviour is expected and reflects the effect of the detection threshold: some signals close to the sensitivity boundary still produce tracks that are well fitted by the inspiral model and therefore yield accurate chirp-mass estimates, although their statistical significance is not sufficient to satisfy the selection criteria. The effect remains small, indicating that the sensitivity threshold is closely aligned with the onset of degraded parameter reconstruction.

IV. CONCLUSIONS

Primordial black hole binaries, in the planetary-to-substellar mass range, have recently attracted renewed interest due to several observational evidences. Detecting such systems using gravitational waves remains challenging, as their long-lasting inspiral signals lie in the intermediate regime between conventional CBCs and CWs searches.

In this work, we have developed and validated a fully operational search pipeline targeting these compact binaries during their inspiral phase. The method is based on the Viterbi algorithm, which recovers the most probable track from the data, assuming that it evolves as a Markov process. To enhance its performance, we introduce a new frequency-domain representation, $(t, f^{-8/3})$, where inspiral chirps become approximately linear, allowing us

to constrain the transition matrix and reduce the signal evolution to a single parameter, the binary’s chirp mass. The recovered Viterbi track is then processed through a candidate isolation procedure, where we extract a potential signal from background noise. Then, the candidate is evaluated using the detection statistics (n_σ , NMSE), which quantify its significance and consistency with the expected inspiral morphology, discriminating possible signal candidates between random noise fluctuations.

Having presented the foundations of the pipeline, we validated the search method using nearly 1000 hours of O3b LIGO Hanford data. After characterizing the background distribution and defining a detection threshold corresponding to a false-alarm ratio of $\text{FAR} = 3\%$, we carried out a search over an injected signal population. We presented the resulting sensitivity curve, specifically the $d_{L,95\%}$. It reaches Galactic scales over most of the explored mass range, extending up to luminosity distances of 135 kpc in the most favorable region. An interesting trend was observed in the high-mass regime, where fast-evolving binaries lead to shorter transients that are more challenging to recover. Nevertheless, the pipeline still performs relatively well, reaching distances $\gtrsim 65$ kpc.

Furthermore, we showed that the NMSE detection statistic provides a useful first estimation of the binary’s chirp mass. As it is has been shown in Fig. 7, within the sensitivity region of the search, the recovered values remain very close to the injected masses, enabling a first characterization of future signal candidates.

The search pipeline presented here is computationally efficient, with the generation of the SFTs representing the main bottleneck. Once the spectrograms are produced, a complete candidate evaluation requires only $\mathcal{O}(\text{sec})$ for a 10-hour data realization. Moreover, we expect a significant increase in sensitivity when using multi-detector data, where uncorrelated noise will help to differentiate between local detector glitches from possible candidates.

Future developments may include leading order corrections to account for eccentricity, a likely property of PBH binaries in this mass range. It would also be interesting to perform dedicated searches over narrower regions of the parameter space, increasing the sensitivity and reducing significantly the computational cost due to the constrained T_{SFT} range. Finally, a systematic search over large datasets of the O3 and O4 runs will be necessary to further investigate the nature of planetary-to-substellar mass PBH binaries, whose discovery could have profound implications for our understanding of dark matter, cosmology, and fundamental physics.

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Appendix A: SNR for long inspirals

In this section, we describe how the signal-to-noise ratio (SNR) of the injected long-inspiral signals is computed. As shown in Fig. 8, the injections span a broad range of SNR values, depending on the source parameters, detector response and noise realization. Since the analysis is performed on real interferometric data, a realistic estimate of the detector noise power spectral density (PSD) is required.

For this purpose, we construct an average PSD directly from O3b LIGO Hanford data. The strain data correspond to the public GWOSC dataset H1:GWOSC-4KHZ_R1_STRAIN from the O3b observing run [45]. Rather than using an analytical sensitivity curve, we estimate the noise spectrum from the same dataset used throughout the analysis. This allows the SNR calculation to account for the actual noise properties of the detector, including spectral lines and deviations from ideal stationary Gaussian noise.

For each O3b strain realization, we estimate the PSD using Welch's method. The strain time series is divided into segments of duration $T_{\text{seg}} = 512$ s with a 50% overlap,

corresponding to a stride of 256 s. A Hann window is applied to each segment before computing its discrete Fourier transform. The one-sided PSD of each segment is then obtained from the corresponding periodogram. To reduce the impact of glitches and narrow spectral artifacts, the individual periodograms are combined using the median estimator rather than the arithmetic mean.

The resulting PSD estimate for each realization, $\hat{S}^{(m)}(f)$, is then averaged over all available O3b segments,

$$\bar{S}(f) = \frac{1}{M} \sum_{m=1}^M \hat{S}^{(m)}(f), \quad (\text{A1})$$

where M denotes the total number of independent noise realizations. This ensemble average provides a robust estimate of the detector noise spectrum used throughout the analysis.

Figure 8 shows the resulting average PSD together with the analytical PSD (aLIGO03LowT1800545), the expected sensitivity for the Hanford detector during O3. The agreement is generally good across the frequency range of interest, while the real-data PSD naturally retains the spectral lines present in the detector data and an expected shift respect to the ideal behaviour. For this reason, we use $\bar{S}(f)$ in all SNR calculations presented in this work.

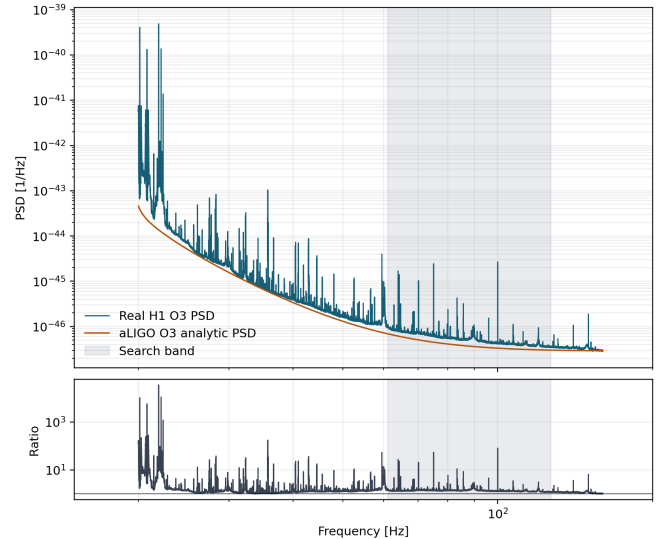


FIG. 8. Average noise power spectral density obtained from O3b Hanford data. We compute the PSD with 1000 hours of O3b data using Welch's method (blue line), compared with the analytic detector sensitivity curve aLIGO03LowT1800545 (orange line). The real data PSD preserves spectral structures and instrumental lines present in the interferometer data.

Once the detector-frame waveform has been generated, the signal is projected onto the Hanford interferometer according to

$$h(t) = F_+ h_+(t) + F_\times h_\times(t), \quad (\text{A2})$$

where F_+ and $F_\times$ are the detector antenna pattern functions [47]. The corresponding Fourier-domain waveform

$\tilde{h}(f)$ is then used to compute the optimal matched-filter SNR.

For a single data frame, the optimal SNR is defined as

$$\rho_{\text{frame}}^2 = \langle h|h \rangle = 4 \int_{f_{\text{low}}}^{f_{\text{high}}} \frac{|\tilde{h}(f)|^2}{\tilde{S}(f)} df, \quad (\text{A3})$$

where the integration is restricted to the optimal frequency interval derived in Sec. II C, [61.1, 126.8] Hz.

In practice, this integral is evaluated numerically on the discrete frequency grid of the waveform, after linearly interpolating the PSD to the corresponding frequency bins. Since the inspiral signal extends over several consecutive data frames, the total SNR is obtained by incoherently

summing the SNR squared from all analyzed frames,

$$\rho_{\text{total}}^2 = \sum_{k=1}^{N_{\text{frames}}} \rho_k^2, \quad (\text{A4})$$

with $N_{\text{frames}} = 8$ in the benchmark searches presented here.

The final optimal SNR reported throughout the paper is therefore

$$\rho_{\text{opt}} = \sqrt{\rho_{\text{total}}^2} \quad (\text{A5})$$

This quantity is used in Fig. 8 to characterize the injected population and to relate the recovered detection statistics to the intrinsic strength of the signal.

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